Hsin Jee Sum

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Personal portfolio: https://hsinjee.vercel.app/

EDUCATION

University of Leeds **Sep 2022 – Jul 2026**
MEng, BSc Computer Science with Artificial Intelligence (Integrated Masters)

- *Academics*: BSc (First Class Honors), MEng (Predicted: First Class Honors)
- *Honors & Awards*: **Microsoft Azure** Certification
- *Relevant Experience*: Custom CNN model for Breast Cancer Classification (Dissertation), AI dev team (Formula Student)
- *Relevant Modules*: Web Services and Web Data – 90%, Robotics – 84%, Artificial Intelligence – 70%

WORK EXPERIENCE

Leeds Gryphon Racing, Formula Student | *C++, Python, ROS2, OpenCV, Gazebo, Computer Vision* ***Leeds, United Kingdom***
Engineering Team; AI R&D Division ***Team Leader*** **Sep 2025 – Present**

- Led a team of 10+ developers to build **ROS2** nodes for sensor fusion (**LiDAR**, camera (computer vision), IMU)
- Built YOLOv8 vision system achieving 92% mAP and 10% faster inference (sub-20ms) for real-time object detection
- Deployed Gazebo simulations that decreased on-track testing hours by 30% through virtual strategy validation
- Optimized FEA heat simulations, achieving a 20% improvement in thermal prediction accuracy for vehicle components

SRKK Group | *Microsoft Azure, Python, Copilot Studio, Git, Document Analysis, NLP, RAG* **Kuala Lumpur, Malaysia**
Summer Intern; Software Development (AI R&D) **Aug 2025 – Sep 2025**

- Investigated **RAG** architectures, improving document retrieval precision by **25%** for **500+** page financial RFP documents
- Developed evaluation scripts that reduced LLM hallucination rates by **18%** through automated faithfulness scoring
- Architected Azure AI workflows, boosting context-window efficiency by **40%** via recursive character text splitting
- Refined prompt strategies, increasing task-specific accuracy by **12%** across **5** distinct corporate reasoning use cases

EH Integrated Systems | *HTML, CSS, JavaScript, MySQL, Docker, C#, Agile methodologies* ***Kuala Lumpur, Malaysia***
Summer Intern; Full-Stack Engineer **Jul 2024 – Sep 2024**

- Collaborated with an **Agile** team of 4 developers to deliver **full-stack** components for a production-ready payment platform
- Designed and deployed containerized APIs with **Docker**, enforcing **dependency inversion** and **clear service boundaries**
- Resolved a database bottleneck by optimizing **SQL** queries and adding indexes, improving data retrieval speed by **30%**

PROJECTS

Agentic RAG for Financial Compliance | *Python, LangGraph, LLMs, Vector Databases, NLP*

- Engineered a multi-agent **RAG** system to automate complex reasoning and compliance checks across dense SEC filings
- Architected a "Critic-Evaluator" loop using LangGraph to identify and mitigate hallucinations in retrieved data
- Optimized retrieval precision by **40%** through iterative agentic feedback loops and semantic conflict resolution
- Synthesized high-fidelity compliance reports by orchestrating specialized agents for data extraction and analysis

Privacy-Preserving ML via Federated Learning | *Python, PySyft, Differential Privacy, Cryptography*

- Developed a Federated Learning framework to train a custom CNN on decentralized data without raw signal exposure
- Implemented Differential Privacy by injecting Gaussian noise into gradients to ensure ε-differential privacy
- Analyzed the privacy-utility trade-off, evaluating the impact of varying noise multipliers on model convergence
- Mitigated data leakage risks in distributed settings, maintaining **82%** accuracy while ensuring formal privacy

Custom CNN model for Breast Cancer Histopathology Images | *Python, TensorFlow, PyTorch, Git, NumPy, Computer Vision*

- Engineered custom CNN in NumPy for breast cancer histopathology to analyze backpropagation and gradient dynamics
- Authored 40-page paper on mathematical derivations and architectural stability for medical image classification
- Evaluated diagnostic robustness via 5-fold cross-validation and F1-score analysis on histopathology datasets
- Optimized for clinical inference, cutting size by **90%** and boosting speed by **80%** while maintaining **85%+** accuracy

TECHNICAL SKILLS

Languages: C, C++, C#, Python, Java, JavaScript
Developer Tools: VS Code
Technologies/Frameworks: HTML, CSS, Django, React, Docker, Git, Azure, ROS2
Libraries: TensorFlow, PyTorch, scikit-learn, NumPy, Pandas, MLflow (familiar), YOLOv8, PySyft
Databases: MySQL, SQLite

EXTRACURRICULAR ACTIVITIES, SKILLS & INTERESTS

Durham University Hackathon, *DurHack 2025 – G Research Prediction Challenge, Honorable Mention*
Leeds University Hackathon, *LeedsHack 2025/26*
Languages: Proficient in English, Cantonese, Mandarin and Malay